

Least Squares

Theory Questions

TTK4260

Question 1 (*Statistical Foundations*)

Consider a dataset with observations $\{y_1, y_2, \dots, y_n\}$.

- (a) Define the sample mean, sample variance, and explain their roles in least squares estimation. [6 points]
- (b) Explain why we square the residuals instead of using absolute values in the least squares formulation. [4 points]

Question 2 (*Least Squares Problem Formulation*)

Consider the linear regression model $y_i = \theta_0 + \theta_1 x_i + \varepsilon_i$ for $i = 1, 2, \dots, n$.

- (a) Write the cost function $J(\boldsymbol{\theta})$ that least squares minimizes. [3 points]
- (b) Express this problem in matrix form $\mathbf{y} = \mathbf{X}\boldsymbol{\theta} + \boldsymbol{\varepsilon}$ and define each component. [6 points]
- (c) What is the geometric interpretation of the least squares solution? [5 points]

Question 3 (*Normal Equations Derivation*)

Derive the normal equations for the least squares problem from first principles.

- (a) Start with the objective function $J(\boldsymbol{\theta}) = \|\mathbf{y} - \mathbf{X}\boldsymbol{\theta}\|_2^2$ and expand it fully. [6 points]
- (b) Take the gradient $\nabla_{\boldsymbol{\theta}} J(\boldsymbol{\theta})$ and set it equal to zero. [8 points]
- (c) Derive the closed-form solution $\hat{\boldsymbol{\theta}} = (\mathbf{X}^T \mathbf{X})^{-1} \mathbf{X}^T \mathbf{y}$. [6 points]
- (d) Under what conditions does this solution exist and is it unique? [4 points]

Question 4 (*Gradient Descent Theory*)

Consider using gradient descent to solve a least squares problem.

- (a) Write the gradient descent update rule for the parameter vector $\boldsymbol{\theta}$. [4 points]
- (b) Explain the difference between batch gradient descent and stochastic gradient descent. [6 points]
- (c) Why is feature scaling important for gradient descent? Provide a mathematical explanation. [6 points]
- (d) What factors affect the convergence rate of gradient descent in least squares problems? [4 points]

Question 5 (*Optimization Methods Comparison*)

Compare different approaches for solving least squares problems.

- (a) Describe the brute force (grid search) approach. When might this be useful despite its computational cost? [5 points]
- (b) Compare the advantages and disadvantages of analytical solutions (normal equations) versus iterative methods (gradient descent). [8 points]

Question 6 (*Non-linear Least Squares*)

Consider a model that is non-linear in its parameters: $f(x, \boldsymbol{\theta}) = \theta_1(1 - e^{-\theta_2 x})$.

- (a) What optimization methods are suitable for non-linear least squares? [6 points]
- (b) How would you implement parameter estimation for this model using gradient-based methods? [7 points]
- (c) What challenges might arise during optimization and how would you address them? [6 points]

Question 7 (*Weighted Least Squares*)

Weighted least squares minimizes $J(\boldsymbol{\theta}) = \frac{1}{2m} \sum_{i=1}^m w_i (y_i - \hat{y}_i(\boldsymbol{\theta}))^2$.

- (a) When and why would you use weighted least squares instead of ordinary least squares? [6 points]
- (b) Derive the weighted normal equations $\mathbf{X}^T \mathbf{W} \mathbf{X} \boldsymbol{\theta} = \mathbf{X}^T \mathbf{W} \mathbf{y}$ where $\mathbf{W}$ is the weight matrix. [8 points]
- (c) How do you choose appropriate weights for different scenarios? Give two examples. [6 points]

Question 8 (*Feature Engineering*)

Feature engineering involves transforming input variables to improve model performance.

- (a) How can you use least squares to fit polynomial models like $y = \theta_0 + \theta_1 x + \theta_2 x^2 + \varepsilon$? [5 points]
- (b) What is the design matrix for fitting $y = \theta_0 + \theta_1 \sin(x) + \theta_2 \cos(x) + \varepsilon$? [5 points]
- (c) Explain why these remain "linear" models from a least squares perspective. [4 points]